

1 Introduction

Periodontal diseases are highly prevalent chronic inflammatory conditions associated with systemic disorders, including cardiovascular disease, metabolic syndrome, and neurodegenerative diseases (1–3). Disease initiation is driven by dysbiotic biofilms and a dysregulated host immune response, leading from reversible gingivitis to irreversible tissue destruction in periodontitis (4–6).

While effective oral hygiene prevents gingival inflammation, active periodontitis requires professional subgingival instrumentation and lifelong supportive care (7, 8). The EFP S3 clinical guideline recommends a stepwise approach including mechanical biofilm control, risk-factor management, subgingival instrumentation, surgical intervention when needed, and supportive maintenance (7). The endpoint is pocket closure (≤ 4 mm probing depth and no bleeding on probing) (9).

A central challenge in periodontal therapy lies in the persistence and tissue invasion of periodontitis-associated bacteria supporting biofilm formation (6). Such biofilms show markedly enhanced tolerance—up to 1,000-fold—to antimicrobial agents compared with planktonic bacteria (10, 11). Consequently, adjunctive antimicrobial strategies have been explored. Systemic antibiotics (typically metronidazole + amoxicillin) are reserved for defined indications due to resistance concerns (12), while topical antiseptics such as chlorhexidine (CHX) show only modest additional benefit (13) and can contribute to antimicrobial resistance and dysbiosis (14, 15).

These limitations have prompted interest in herbal medicines which combine affordability, high biocompatibility, and favorable safety (16). Among the phytotherapeutics, *Nigella sativa* (black seed) and its leading compound thymoquinone (TQ) display potent antioxidant, antimicrobial, anti-inflammatory, and bone-regenerative actions (17–20). This systematic review summarizes and critically assesses the evidence for the clinical effectiveness of preparations from *Nigella sativa* in periodontal therapy.

2 Methods

Following peer-review, the original PubMed search was expanded to include Embase, Scopus, Web of Science and Cochrane Central Register of Controlled Trials (CENTRAL). The systematic review was subsequently conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (21) statement using a rigorous methodology to minimize the risk of bias. Ethical approval was not required because all data were obtained from previously published studies.

The final search of all databases was completed on July 16, 2026; therefore, publications indexed after this date were not eligible for inclusion. The search strategy combined terms related to *Nigella sativa* and its active constituent TQ with terms describing gingivitis, oral mucositis, periodontitis, and peri-implantitis. The following search string was adapted for each database: (“*Nigella sativa*” OR “black cumin” OR “black cumin seed” OR “black seed” OR kalonji OR thymoquinone) AND (periodontitis OR gingivitis OR “oral mucositis” OR stomatitis

OR “periodontal disease*” OR “oral disease*” OR “oral inflammation” OR “oral health”). Eligible studies included randomized controlled trials, randomized clinical trials, controlled clinical trials, and prospective clinical studies published in English and involving human participants. Exclusion criteria comprised *in vitro* studies, animal studies, narrative or systematic reviews, case reports or case series, studies investigating aggressive periodontitis (Grade C periodontitis) or multicomponent interventions in which the independent effect of *Nigella sativa* or TQ could not be determined, and studies outside the predefined disease spectrum of the present review.

Following the initial search, all the retrieved results were imported into Excel for extraction. All articles where a *Nigella sativa* preparation was either used alone or compared to any other intervention for gingivitis/mucositis or peri-implantitis-periodontitis were included. Duplicates were removed before title and abstract screening. Two reviewers (SCH and AA) independently performed title/abstract screening and full-text assessment according to the predefined eligibility criteria. Discrepancies at the full-text stage were resolved through discussion and consultation with a third reviewer (JV). Data were extracted using a customized data extraction form. The PICO framework was used to structure the focus of this literature review, ensuring a comprehensive and systematic approach to evaluating clinical studies where *Nigella sativa* preparations were used to improve oral health. The PICO components were defined as follows:

P: Patients—The review included patients of all ages with gingivitis/mucositis and peri-implantitis/periodontitis.

I: Intervention—The interventions considered were preparations from *Nigella sativa* seed.

C: Comparison—Comparisons were made between the groups (placebo or active control).

O: Outcome—The primary outcome was improvement in gingival, mucosal, and periodontal health assessed using established outcome measures appropriate for the condition investigated, including clinical indices (e.g., PI, GI, BOP, PPD, CAL, OMAS, WHO mucositis grade), patient-reported outcomes (e.g., VAS pain and swallowing), and biochemical markers (e.g., IL-6, TNF- α , MMP-8, TAC, ALP). Data were extracted using a customized data extraction form. For the purpose of this review, the following information was extracted and tabulated for all the selected studies: title; first author’s name; study design, publication year; study location, population description; intervention, frequency, mg TQ per dose, comparator, established outcome measure(s), follow-up period, and overall RoB. A meta-analysis was not performed due to the significant clinical and methodological heterogeneity between studies.

3 Clinical evidence

3.1 Overview of search

Figure 1 presents the PRISMA 2020 flow diagram of the study selection process. The detailed search strategies are presented in